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# Transformer with multi-head attention mechanism for bearing fault detection

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ABSTRACT

Timely fault detection of bearings is vital to ensure the uninterrupted functioning of rotating machinery in industry. Research on fault detection strategy for bearings has accelerated due to developments in artificial intelligence and computation, although managing large amounts of intricate sensor data remains complex. The traditional approaches rely on handcrafted features and shallow learning models, which in high-dimensional sensor data, may struggle to capture complex fault patterns. Using the transformer model with a multi-head attention mechanism, this work designs an innovative bearing fault detection method. Transformer architecture, renowned for its efficacy in natural language processing applications, is tailored to handle sequential sensor input efficiently, without requiring identifiable features. The multi-head attention mechanism enables the model to capture both local and global dependencies that are essential for fault recognition, which aids in the model's focus on various input sequence segments. The model was evaluated on a two real-world benchmark bearing dataset namely, Case Western Reserve University (CWRU) bearing dataset and Society for Machinery Failure Prevention Technology (MFPT) bearing dataset and compared with the deep learning baselines. This work also eliminated the need for extensive data processing and achieved superior fault detection accuracy. The results reveal the efficacy of the transformer model with multi-head attention in accurately detecting bearing faults, showcasing its potential for real-time condition monitoring in industrial scenarios. These findings demonstrate transformers' contribution to predictive maintenance advancement, guiding the industry toward scalable, real-time monitoring that reduces downtime and increases sustainability. The future work could integrate the edge computing for on-device use and adapt to multi-fault scenarios.

## 1. Introduction

Bearings are prime movers of rotating machines, like motors, turbines, generators, and other mechanical systems. The bearing is also known as a machine element that lessens friction between moving elements and restricts relative motion to only the desired motion. The essential parts of the bearing include rings (outer and inner race), rolling elements (balls, rollers, needles), cage, and fitting surfaces. The different bearing faults include outer race fault, inner race fault, ball defect, and cage defect [1]. Bearing health directly affects the overall health of the machinery and can directly affect performance and safety. Early detection of bearing faults is vital to ensure smooth operation of the machine, preventing catastrophic failure while allowing efficient scheduled maintenance. The traditional bearing fault detection (BFD) approaches are based on manual feature extraction from vibration signals, followed by classification using statistical methods and machine learning (ML) methods. However, manual feature extraction is a labor-intensive task that is prone to human error. It also requires a great deal of experience and prior knowledge. Algorithms, like support vector machines (SVMs), decision trees (DTs), random forest (RF), and k-nearest neighbor (kNN), have been extensively used for the BFD [2–5]. Using these ML techniques presents several challenges, including the need for effective feature engineering. SVM is also sensitive to the choice of the kernel function and its parameters, which can significantly impact the BFD performance. Also, the performance of SVM suffers under noisy data or high-dimensional data. DT is also prone to overfitting when dealing with complex or noisy data. A small variation in the training data can lead to very different tree structures, resulting in instability. Although RFs are

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more resilient than DTs, tuning them can be challenging, due to the numerous trees and features that need to be considered at each split. When dealing with large datasets, kNN is computationally expensive, as it requires calculation of the distance between the test sample and all training samples. Also, as the concept of nearest neighbors becomes less meaningful, it may not perform well in high-dimensional feature spaces [6]. These algorithms have been employed for the BFD, but they have their own constraints and weaknesses. Recent years have seen widespread use of deep learning (DL) for the BFD. Algorithms like convolutional neural networks (CNN), deep Boltzmann machines (DBM), stacked auto-encoders (SAE), and numerous others have been extensively used in the fault diagnosis domain. Without the need for manual feature engineering, DL models can automatically identify and extract pertinent characteristics from unprocessed vibration signals. A significant flaw in ML algorithms is naturally addressed by DL models [7]. DL models have also shown better FD accuracy, compared to classical ML models like SVMs and RFs. Also, DL models can handle large and complex datasets, making them more scalable than conventional ML approaches [8]. This simplifies the overall fault detection process and leads to better performance.

In recent times, much emphasis has been placed on DL application in the BFD to where DL has become a cornerstone of BFD domain research. CNN has been among the most dominant algorithms in FD applications, and has also yielded good results. Among DL models, CNN has been among the most popular choices in the domain of the BFD. CNN can use the vibration data to perform autonomous feature engineering. It can also handle large and complex datasets, and aid in developing efficient BFD methods. Reference [9] used CNN in combination with the spectrogram of the vibration signals for various BFDs. In Ref. [10], BFD techniques were developed with the help of CNN, having a rectangular input shape and rectangular convolutional kernels. The physics-guided CNN has also been shown to perform better than the standard CNN models. Reference [11] proposed a BFD method that employed spectrogram images as an input to the CNN model with switchable normalization; the proposed method was assessed on the Case Western Reserve University (CWRU) public domain bearing dataset. Reference [12] employed variational mode decomposition to eliminate noise from the motor signals, and Mel spectrum voiceprint features were extracted; CNN layers were utilized to extract these features, and the Mel–CNN model was applied for the BFD. Reference [13] proposed a BFD utilizing the representation learning method based on spatiotemporal tensors and the CNN model. Reference [14] proposed a multiscale fully CNN model for the BFD under a noisy environment; after retrofitting, the vibration signals were used in convolution with a sizable kernel to extract temporal features across a wide area, and suppress high-frequency noise. Reference [15] demonstrated the application of CNNs in extracting local features from vibration signals for the accurate BFD. CNNs are excellent at identifying spatial details in images. On the other hand, they have trouble capturing long-term dependencies in the sequence of 1D signals. In general, CNNs are primarily designed to capture localized features within a limited window of the signal. This can be a limitation for the BFD, where fault-related information might span over larger sequences in the vibration data. In recent times, vision transformers (ViTs), a DL architecture proposed by Vaswani et al. [16], have revolutionized natural language processing tasks. Transformers are rapidly being used in various instances of time-series anomaly detection, in addition to their widespread use in natural language processing. Anomaly transformers can outperform baseline anomaly detection techniques, raising F1 scores by up to 17% and lowering training times significantly, as shown by extensive trials on benchmark datasets [17]. In addition, by combining reversible instance normalization with the anomaly transformer architecture, the reversible instance normalized anomaly transformer proposed in Ref. [18] improves the capacity to identify anomalies in time series data. The attention mechanism used by transformers enables the model to concentrate on particular segments of the input sequence that are most pertinent to the task at hand. A technique known as self-attention enables each element in an input sequence to focus on every other element, thereby capturing contextual linkages throughout the sequence. The ability of the model to capture a variety of relationships in complicated signals may be limited by single-head attention, a particular type of self-attention that employs a single set of queries, keys, and values. On the other hand, multi-head attention (MHA) allows the model to focus on numerous representation subspaces at once by using various sets of queries, keys, and values in parallel. This mechanism in transformers provides the advantage of capturing diverse representations, enhanced feature extraction, and improved model capacity [19–21]. This inherent capability of transformers makes them well suited to analyzing time series signals, such as in the case of BFD, using vibration signals, where capturing long-range dependencies is crucial. Inner race defects, outer race defects, ball defects, cage defects, and combined or mixed faults (e.g., simultaneous damage in inner and outer raceways) are the most commonly described anomalies in the literature. Each of these fault types contributes to the development of deterioration and modifies the bearing's vibration characteristics in unique ways. It is crucial to carefully specify the area of research because not all of these abnormalities are regularly available or well-represented in benchmark datasets. In the proposed study, for the CWRU bearing dataset, inner race, outer race, and ball defects are taken into consideration in this study and for the Society for Machinery Failure Prevention Technology (MFPT) bearing dataset [25], inner race and outer race defects are taken into consideration. There are two reasons behind this choice. First, the MFPT dataset only contains inner and outer race faults in adequate quality and quantity, but the CWRU dataset offers well-labeled and validated samples for all three defect categories. The second point of practical significance is that the most important and common fault modes in industrial bearings are inner and outer race defects, which immediately lead to system failure and degradation. In order to expand the scope of the evaluation, ball defects which are less frequent in practice were kept for the CWRU dataset, whereas cage and combination faults were eliminated because of their inconsistent annotation and low representation.

Inspired by the above features of the transformers and MHA mechanism, this work proposes a data-driven BFD method using the transformers with the MHA mechanism to address the shortcomings of the CNN-based BFD approaches. The key contributions of the work are:

1) A transformer backed with an MHA mechanism is introduced for various FDs in bearings. This scheme better captures the complex and non-linear fault patterns in the vibration signals, compared to the local feature extraction of the CNNs.
2) The proposed model can apprehend distant relations in the data, which is beneficial for FD. The long- and short-range temporal relationships of the vibration signals are also captured.
3) The efficacy of the proposed approach in actual industrial settings is demonstrated by evaluating its performance in noisy environments.
4) The effectiveness of the proposed strategy is demonstrated through in-depth analysis in CWRU and MFPT bearing dataset, and a comparison made with baseline methods. The findings show that the suggested methodology facilitates effective BFD.

There are several sections to the remaining portion of this work. Section 2 describes the preliminary design of the transformer with an MHA mechanism. Section 3 covers the specifics of the suggested methodology, while Section 4 provides specifics of the experimental configuration, data collection process, results and a discussion of the proposed work, while Section 5 concludes the work.

## 2. Transformer with a multi-head attention mechanism

The transformer belongs to the category of DL architecture and has obtained substantial consideration in natural language processing and computer vision tasks. It relies on the self-attention mechanism (SAM)

instead of recurrent or convolutional layers, making it highly parallelizable, and enabling it to more effectively capture long-range dependencies [16]. Transformers can learn long-range relationships by attending to entire sequences, unlike recurrent neural networks, which repeatedly process sequence elements, and that are restricted to short-term dependencies. The transformer model comprises a self-attention mechanism, MHA, and a position-wise feed-forward network (PWFFN) integrated into a transformer encoder [22]. With an input sequence of the form $[x_1, x_2, x_3, \ldots, x_t]$, where $t$ represents the position of each instance, the input to the transformer model is often in the form of the tensor $\mathbb{R}^b \times \mathbb{R}^n$, where $b$ and $n$ are the batch size and sequence length, respectively. The input to the transformer is first processed via an embedding layer, which transforms it into a one-hot token of dimensions $\mathbb{R}^b \times \mathbb{R}^n \times \mathbb{R}^d$, where $d$ is the number of embedding dimensions. Next, the newly computed tensor is combined with a position encoding of sinusoidal form, and fed through an MHA module. A normalization layer is used before and after the self-attention mechanism with a residual connection. Subsequently, the resulting output is directed to a two-layer PWFFN. Similar to the previous step, this network also joins the input and output using a residual connection, followed by another normalization layer. Fig. 1 shows the entire transformer model architecture, while the following sections describe each component.

### 2.1. Self-attention mechanism

Self-attention is the fundamental module of the transformer model, with the purpose of centering on certain regions of the input data to extract meaningful feature information. In classification tasks, the self-attention mechanism learns the discriminative feature from the data, making it better in terms of extracting the internal correlation of the data, rather than relying on external information [23]. SAM receives the input embedding $X$ of dimension $d$ from the embedding layer, and applies a linear transformation. This linear transformation converts the embeddings into queries (Qs), keys (Ks), and values (Vs). Then, the dot product of Q and K is computed, resulting in a single-head scaled dot-product attention. After dividing it by a scaling factor $\sqrt{d_k}$ to obtain the corresponding weight values, a SoftMax activation function is then applied to their dot product. This transformation is expressed as:

$$A_s(Q_s, K_s, V_s) = softmax\left(\frac{Q_s K_s^T}{\sqrt{d_k}}\right) V_s \tag{1}$$

where, $\mathrm{Q_s} = XW_s^q$, $\mathrm{K_s} = XW_s^k$, and $\mathrm{V_s} = XW_s^v$, with dimensions $d_q$, $d_k$, and $d_v$.

### 2.2. Multi-head attention

Multi-head attention is the extended form of the SAM that permits the transformer model to pay attention to various sub-sequences. Therefore, instead of mapping with a single set of linear transformations, the input tokens are transformed into $h$ queries, keys, and values. This results in $h$ different outputs through parallel self-attentions. These individual outputs ($h_i$) are then concatenated and re-projected, forming an MHA mechanism as expressed in the following equations:

$$A_h(X) = concat(h_1, h_2, h_3, \ldots, h_i)W^0 \tag{2}$$

$$h_i = A_s(XW_i^q, XW_i^q, XW_i^q) \tag{3}$$

where, the $\mathrm{i}^{\mathrm{th}}$ projection on the embedding $X$ is used to find queries, keys, and values that are denoted as: $W_\mathrm{i}^q \in \mathbb{R}^{d \times d_q}$, $W_\mathrm{i}^k \in \mathbb{R}^{d \times d_k}$, and $W_\mathrm{i}^v \in \mathbb{R}^{d \times d_v}$, respectively. Moreover, $\mathrm{W}^0 \in \mathbb{R}^{h.d_v \times d}$ represents the linear projection on the concatenated multi-head, while $d_k = d_v = d/h$ in the transformer model.

### 2.3. Position-wise feedforward networks

A two-layer position-wise feed-forward network receives the output from the MHA module. The hidden layers in this network are activated using the Rectified Linear Unit (ReLU) activation function. The feed-forward layer independently processes each position, therefore making it a PWFFN. The operation of the PWFFN is expressed as:

$$FFN(X_A) = ReLU(0, X_A W_1 + b_1)W_1 + b_2 \tag{4}$$

where, W and b denote the weights and bias of the layers of the network, d denotes the number of embedding dimensions, and $\mathrm{W_1} \in \mathbb{R}^{d \times d_{FFN}}$, $\mathrm{W_2} \in \mathbb{R}^{d_{FFN} \times d}$, $\mathrm{b_1} \in \mathbb{R}^{d_{FFN}}$, and $\mathrm{b_2} \in \mathbb{R}^d$.

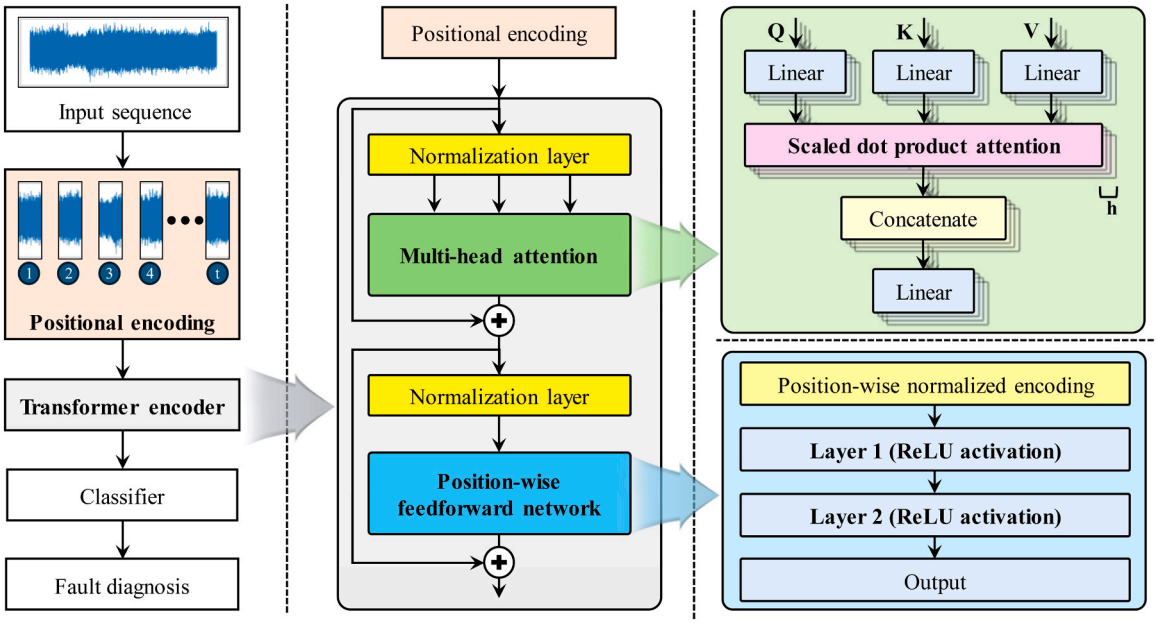



**Fig. 1.** The proposed transformer (MHA-T) model with its components is used for BFD.

### 2.4. Transformer encoder

The transformer encoder is the main module of the transformer architecture, which integrates the attention head mechanism, multi-head attention, and the PWFFN. Within the transformer encoder, both the MHA module and PWFFN use a residual connection and the normalization layer. The combination of all these blocks results in the transformer encoder. While the previous sections have already explained the remaining blocks, the two normalization layers present in the transformer encoder can be expressed as:

$$X_A = Norm(A_h(X) + X) \tag{5}$$

$$X_{FFN} = Norm(FFN(X_A) + X_A) \tag{6}$$

where, $X_A$ and $X_{FFN}$ represent the outputs of the multi-head attention and PWFFN, respectively. Furthermore, it is important to note that while multiple transformer encoders stacked together in the same transformer model share the same architecture, they do not share the same hyperparameters.

## 3. The proposed methodology

In recent times, computer vision tasks have made substantial use of the vision transformer model. Studies show that a vision transformer model based on the attention mechanism can also be directly applied, instead of the convolutional structure. In addition, the vision transformer model has advantages of capturing diverse representations, enhanced feature extraction, and improved model capacity [19–21]. This paper therefore proposes the use of the transformer model with MHA for the BFD. Considering the one-dimensional characteristics of the original vibration data acquired from the electric motor, an end-to-end MHA-based transformer model, namely MHA−T, is proposed for the BFD. This model can work on the direct input of the original vibration data and facilitate autonomous feature extraction. Fig. 1 shows the layout. The original Transformer framework was developed for sequential data, where inputs are represented as a series of tokens. In the proposed study, the Transformer was adapted for 1D time-series vibration signals. Each input signal was segmented into non-overlapping 1D patches of length L, where L was chosen based on the signal sampling rate and the frequency range of interest. A fixed-dimensional embedding was then obtained by flattening each patch and projecting it through a linear layer. These embeddings, along with positional encodings, were provided as input token to the Transformer encoder. The multi-head self-attention mechanism enables the model to capture long-range dependencies in the time-series data, while the patching strategy preserves local temporal patterns. The proposed methodology involves the development of an MHA-based transformer model for multiple fault detection in the bearings. The MHA mechanism is a core component of the proposed transformer architecture that helps the model focus on information pertinent to the task by allowing it to pay attention to particular segments of the input sequence. The structure of MHA−T model can be segregated into different parts, which are given below:

### 3.1. Transformer block

This block contains the "transformer_block" function, which represents a single block within a Transformer architecture. It relies on the SAMs to weigh the significance of numerous input elements against each other, making them highly suitable for sequence-based tasks. It comprises the following layers:

- **Multi-head attention layer:** This layer calculates the attention score between a query (Q) and a key (K) matrix to retrieve values (V). The formula used is:

$$MHA(Q, K, V) = softmax\left(QK^T \Big/ \sqrt{d_k}\right)V \tag{7}$$

where, $d_k$ denotes the key dimension.

- **Layer normalization:** This normalizes the output of the attention layer, and helps improve convergence during training. The normalization formula is:

$$LayerNormalization\,(x) = \quad (x-\mu)/(\sigma + \in) \tag{8}$$

where, $\mu$ and $\sigma$ denote the mean and standard deviation of the input tensor, respectively, and $\in$ denotes a small constant for numerical stability.

- **Feed-forward network (FFN):** The output from the normalization passes through this network with a ReLU activation function:

$$FFN(x) = ReLU(W_1 x + b_1) \bullet W_2 + b_2 \tag{9}$$

where, $W_1$ and $W_2$ denote the weight matrices, and $b_1$ and $b_2$ represent the bias vectors.

- **Residual connection:** A residual link is formed by adding the original input tensor to the output of the feedforward network. This aids in preventing the training-related vanishing gradient problem.

### 3.2. Transformer module

This module assembles the overall model architecture, incorporating the transformer block to learn the hierarchical representations. It comprises dense layers, instances of transformer block, global average pooling (GAP) layers, and a dense layer with a SoftMax activation function. Using a dense layer, the input features are first converted into an embedding space. This layer learns the first representation of the input data. There are successive stacks of "transformer_block" functions. Employing self-attention, normalization, and feedforward processing, every block enhances the input features. A GAP layer is added after the Transformer blocks to minimize the spatial dimensions of the output features. Feature data from the entire sequence is combined in this layer. To generate class probabilities for multi-class classification tasks, a dense layer with SoftMax activation is added at the end.

## 4. Application on bearing dataset

For the implementation of proposed methodology, the raw vibration signals were segmented using an non-overlapping sliding window approach with a window length of 784 samples and a stride of 300 samples. Each segment was designated with the fault category that matched the window's end. The proposed one-dimensional transformer model initially projects input segments of shape (784, 1) onto a 64-dimensional embedding space. Multi-head self-attention, a feed-forward layer of dimension 64 with ReLU activation, residual connections, and layer normalization ($\varepsilon = 1e{-}6$) are all incorporated in the architecture, which is made up of four stacked Transformer blocks. The number of attention heads was changed between {2, 4, 8} for ablation analysis, and a configuration without multi-head attention was also assessed. Global average pooling is applied before the final SoftMax classification layer. With a batch size of 50 and a maximum of 50 epochs, the model was trained using stochastic gradient descent (learning rate = 0.01) with categorical cross-entropy loss. As a regularization technique, early stopping with a 10 epoch patience based on validation loss was used. All trials were conducted five times using controlled random seeds, and the dataset was divided into 70% training and 30% testing sets with shuffling enabled. The total number of trainable parameters is explicitly reported in the manuscript to ensure full reproducibility.

### 4.1. CWRU bearing dataset

#### 4.1.1. Experimental set-up

The experimental set-up shown in Fig. 2 comprises a 2 hp electric motor, dynamometer, and control electronics (not indicated). The test bearing supports the shaft of the motor. Using electro-discharge machining with varying fault dimensions, single-point faults were induced into the test bearings to emulate the various faults in the bearing. The vibration data were gathered with the help of accelerometers, which were placed in the magnetic base housing. These data were acquired with a sampling frequency of 48 kHz. The different fault conditions include faults in the inner race (FIR), outer race (FOR), and ball bearing (FBB). The data were acquired under different operating conditions, including various loading conditions like (0, 1, 2, and 3) hp and various operating speeds from the drive end of the motor. The data conditioning and model development was on a PC with a Windows 11 system, Intel Core i7 processor, NVIDIA GeForce RTX 3080 graphics card, and 32 GB memory.

For the model assessment, these vibration data were used after data conditioning. Data conditioning is critical to ensure that the model learns the relevant patterns and extracts meaningful attributes using the input data. The signal's amplitude has been normalized between 0 and 1 to stabilize the training, and ensure consistent signal magnitudes across different samples. After the conditioning, the continuous vibration signals were divided into fixed-length segments to create a sequence suitable for input into the proposed transformer model. Then, each segment is treated as a sequence of data points within the segments, with each data point being considered as a token. The tokenized segments with positional encodings are arranged as an input to the transformer with an MHA mechanism. It is worth noting that each segment is a sequence of tokens representing the vibration signal data. It is ensured that all input sequences (segments) are of the same length. After this, the transformer model is designed in such a way that it can accommodate input dimensions, sequence length, and attention mechanisms. The proposed model was trained using the stochastic gradient descent (SGD) optimizer and categorical cross entropy (CCE) loss function. SGD is generally considered better for achieving higher generalization performance. SGD tends to converge to flatter minima, which are more robust, and generalize better to unseen data. To achieve the best possible generalization performance, especially on complex tasks and large datasets, SGD is generally preferred [24]. The CCE loss function was employed for the BFD, as it is a common loss function for multi-class classification problems. Stable gradients are produced during backpropagation by the formulation of CCE loss, which facilitates smoother optimization and convergence during training.

#### 4.1.2. Results

The proposed model has been rigorously evaluated to ensure its optimal performance. The model was trained on the CWRU bearing dataset using the optimum network structure and hyperparameter optimization. During training, a batch size of 50 was used. The trial was repeated 10 times to ensure efficient training and optimal model selection. Figs. 3 and 4, respectively, show the accuracy and loss curves for training and validation. These figures show that following roughly 50 epochs, the model has begun to converge, and the accuracy and loss values for the training and validation datasets have stabilized. It is noteworthy that during the initial phase, the training loss surpasses the validation loss. When used in conjunction with training, dropout reduces the model's capacity, and promotes robust feature learning. The training and validation losses of the network are essentially constant at the same value, which implies that the network has high generalization capabilities.

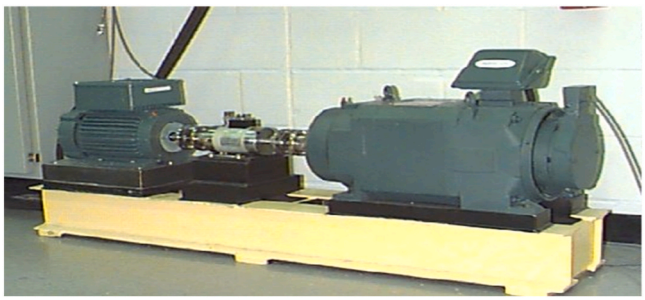


**Fig. 2.** Experimental setup for vibration data acquisition.

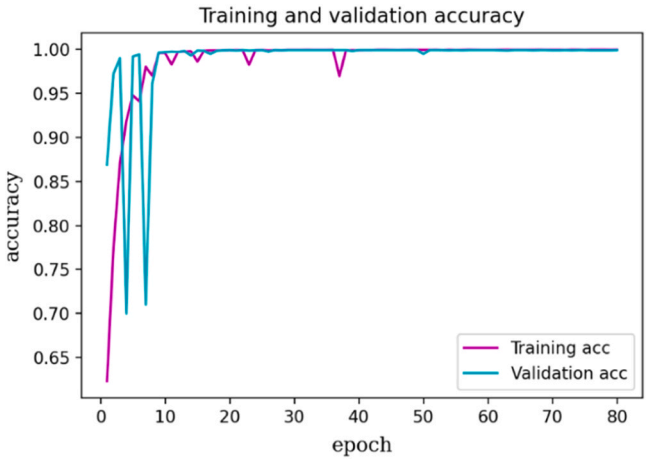



**Fig. 3.** Accuracy curve for the proposed MHA-T model for CWRU dataset.

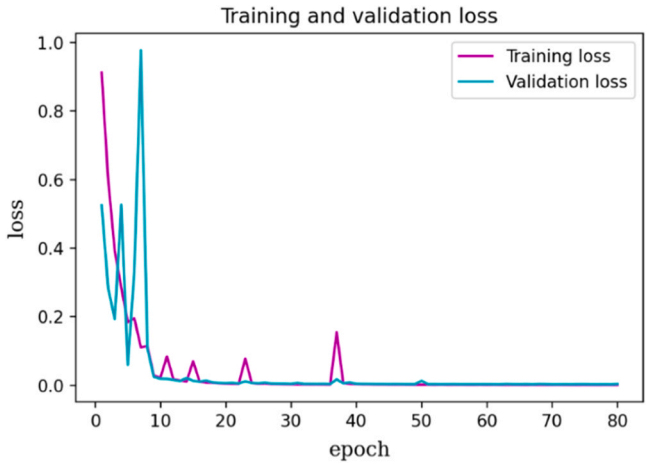



**Fig. 4.** Loss curve for the proposed MHA-T model for CWRU dataset.

To evaluate the performance, the testing dataset is classified using the trained model. Training and tests on the data were repeated multiple times to ensure good performance. Table 1 shows the results. The performance of the model is also compared by computing the total number of trainable parameters and the average training time. Real-time applicability was assessed by measuring inference latency and memory usage. On average, the model required less than 1.85 ms per sample with approximately 2452.87 MB of memory, validating its efficiency for

**Table 1**
Performance details of the MHA-T model for CWRU data.

| Model | Average accuracy (%) | Number of parameters | Training time (s) | Inference time per sample | Memory usage |
|---|---|---|---|---|---|
| MHA-T | 99.52 | 300,164 | 1336.96 | 1.85 ms | 2452.87 MB |

online fault detection. An accuracy of about 99.52% has been attained on average. In order to assess the statistical reliability of the results, each experiment was repeated five times, and the mean accuracy was 99.52% with a 95% confidence interval of 99.10% - 99.94%. A table that shows the performance of a classification model visually is called a confusion matrix (CMX) in the context of DL-based classification; it provides an overview of the number of instances of each class that were false positives and false negatives, respectively, and the number that were correctly predicted (true positives and true negatives). Fig. 5 shows the CMX for the proposed MHA-T model. It is evident from the CMX that all the states of the bearing, namely FIR, FOR, FBB, and N, have been classified effectively. The different performance factors, precision ($p$), sensitivity ($s$), and F1 score, were measured to assess the performance of the MHA-T model. These values were used to estimate performance. These factors can be calculated as follows:

$$p = \frac{tp}{tp + fp} \tag{10}$$

$$s = \frac{tp}{tp + fn} \tag{11}$$

$$F1 = \frac{2 \times p \times s}{p + s} \tag{12}$$

where, $tp$ indicates actual positive trials, $tn$ indicates actual negative trials, $fp$ indicates incorrect positive trials, and $fn$ indicates incorrect negative trials. Table 2 lists the values of $p$, $s$, and F1 scores. These parameter values are good, indicating the efficient performance of the proposed MHA-T model.

#### 4.1.3. Ablation study

To further support the proposed structure, an ablation study was carried out with a focus on the influence of MHA, the number of attention heads, and the segment length of the input sequence. These components are crucial to the design of transformer-based models, and assessing their function helps to understand the rationale behind selecting a structure. A comparison was made between the length of the input segment, the number of attention heads, and the MHA's contribution to the overall model performance. When MHA was removed and a straight skip connection was employed in its stead, accuracy continuously dropped, implying that cross-segment feature interaction is crucial for fault classification. The findings demonstrated that performance increased as the number of attention heads increased to four,

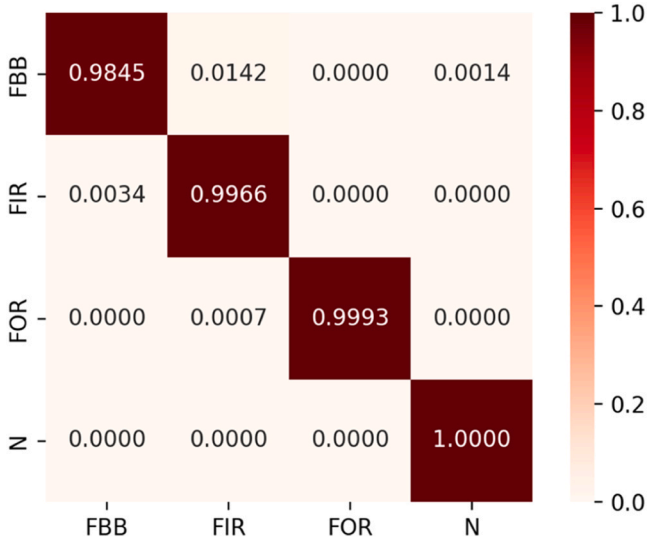



**Fig. 5.** CMX of the proposed MHA-T model for CWRU dataset.

**Table 2**
Performance factors of the MHA-T model for CWRU data.

| State | $p$ | $s$ | F1 score |
|---|---|---|---|
| FBB | 0.996 | 0.984 | 0.989 |
| FIR | 0.985 | 0.996 | 0.990 |
| FOR | 1.0 | 0.999 | 0.999 |
| N | 0.998 | 1.0 | 0.998 |

whereas over-parameterization caused additional increases to have little or even detrimental effects. A range of two and four attention heads were used. The effect of segment length was examined using window sizes of 256, 512, and 784 samples. The 512–784 range yielded the best results, whereas shorter segments provided worse accuracy due to insufficient temporal context, while longer segments enhanced performance by capturing richer temporal dynamics. Table 3 displays the entire ablation results. Overall, the ablation study supports the architectural choices made in the proposed design by demonstrating that using four attention heads, enabling MHA, and implementing moderate segment lengths results in the most efficient arrangement.

#### 4.1.4. Performance evaluation in noisy environments

It is important to study the model's performance in a noisy environment. Adding white Gaussian noise is a popular method that is aimed at emulating noisy environmental conditions, to further examine the performance of the proposed BFD technique. The primary goal of introducing noise is to evaluate the effectiveness of the proposed method, and verify its resilience when processing noisy data. It also offers insights into the performance of the algorithms in harsh environments, and aids in understanding their limitations. Furthermore, extremely noisy surroundings may result in impractical conditions that alter the essence of the issue at hand. As opposed to generalized noisy circumstances, it can force the model to learn particular degradation conditions. More precisely, excessive noise levels might necessitate model tuning that performs poorly in less noisy environments, which could result in overfitting to improbable events, and reduce the diagnostic utility of the predictive mechanism. Therefore, a trade-off exists between difficult, noisy conditions, and unrealistic situations. This is significant when dealing with different faults, and it is crucial to add noise without affecting the severity level of faults. To maintain the ability to distinguish between severity levels, an equilibrium must be reached between realistic and excessive noise, with enough noise added to replicate real-world scenarios. When assessing how well algorithms function in noisy, simulated environments, this is a crucial factor that needs to be considered.

$$\mathrm{SNR} = 10\log\left(\frac{\mathrm{p_{signal}}}{\mathrm{p_{noise}}}\right) \tag{13}$$

where, $p_{signal}$ denotes signal power, and $p_{noise}$ denotes noise power. This research, therefore, adheres to signal-to-noise ratio (SNR) range values

**Table 3**
Ablation study based on MHA, number of heads and segment length on CWRU dataset.

| MHA | Heads | Segment length | Accuracy (%) |
|---|---|---|---|
| TRUE | 2 | 256 | 70.10 |
| | 2 | 512 | 73.20 |
| | 2 | 784 | 83 |
| | 4 | 256 | 78.80 |
| | 4 | 512 | 90.24 |
| | 4 | 784 | 99.24 |
| FALSE | 2 | 256 | 66.34 |
| | 2 | 512 | 71.12 |
| | 2 | 784 | 76.29 |
| | 4 | 256 | 78.34 |
| | 4 | 512 | 82.56 |
| | 4 | 784 | 88.45 |

that have been extensively examined in the field of FD. SNR range levels ranging (−5–10) dB have been considered for the analysis. All the experiments are conducted independently 10 times, with the structure and parameters of the proposed model remaining unchanged. Also, to extend robustness of the proposed approach, Laplace noise was also injected, which better represents heavy-tailored industrial disturbances. This provides a more realistic robustness evaluation. Fig. 6 shows the accuracy of the proposed model under different SNR levels for both noises. The findings show that the classification accuracy of the proposed model generally improves with increasing SNR levels. Higher SNR indicates less noise in the input signal, which makes classification simpler. Depending on the SNR level, the model performs differently. In essence, it is evident that while the model performance can be affected by noisy conditions, the proposed approach has also worked efficiently under noisy conditions.

### 4.2. MFPT dataset

#### 4.2.1. Experimental setup

The Society for Machinery Failure Prevention Technology (MFPT) is the source of the MFPT bearing dataset [25]. A NICE bearing is used in the MFPT's bearing data. Three classes of bearing data are included in this dataset: normal bearing data, inner race fault, and outer race fault at different loads. The test setup was equipped with a NICE, 7 outer race fault conditions: 25, 50, 100, 150, 200, 250 and 300 lbs of load and 7 inner race fault conditions: 0, 50, 100, 150, 200, 250 and 300 lbs of load. For the proposed model assessment, these vibration data were used after data conditioning. A data conditioning and processing strategy similar to CWRU bearing dataset were followed for this dataset. The proposed model was trained using the SGD optimizer and CCE loss function.

#### 4.2.2. Results

In order to ensure optimal performance, the proposed model has undergone a thorough evaluation. Using hyperparameter optimization and the optimal network topology, the model was trained on the MFPT bearing dataset. A batch size of 50 was used for training. To guarantee effective training and ideal model selection, the trial was conducted multiple times. The accuracy and loss curves for training and validation are displayed in Figs. 7 and 8, respectively. According to these figures, the model has started to converge after about 80 epochs, and the accuracy and loss values for the training and validation datasets have leveled out. Dropout encourages strong feature learning while lowering the model's capacity when used in conjunction with training. The network's great generalization capabilities are implied by the fact that its training and validation losses are nearly constant at the same value.

The trained model is used to classify the testing dataset in order to assess performance. To guarantee optimal performance, training and testing on the data were conducted several times. The results are displayed in Table 4. Real-time applicability was assessed by measuring inference latency and memory usage. On average, the model required less than 1.76 ms per sample with approximately 2102.87 MB of memory, validating its efficiency for online fault detection. By calculating the average training time and the total number of trainable parameters, the model's performance is also compared. On average, an accuracy of 97.11% has been achieved. In order to assess the statistical reliability of the results, each experiment was repeated five times, and the mean accuracy was 97.11% with a 95% confidence interval of 96.54% - 97.68%. The CMX for the proposed model is displayed in Fig. 9. The CMX makes it clear that every bearing state, FIR, FOR, and N has been successfully categorized. Table 5 lists the values of *p*, *s*, and F1 scores. The different performance factors were measured to assess the performance of the proposed model. Performance was estimated using these parameters.

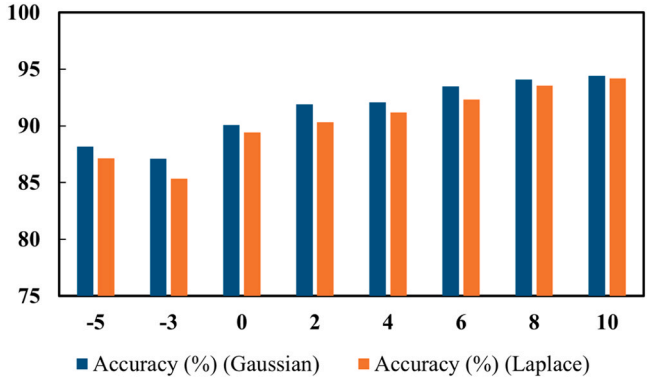



**Fig. 6.** Performance of the proposed method under different SNRs for CWRU data.

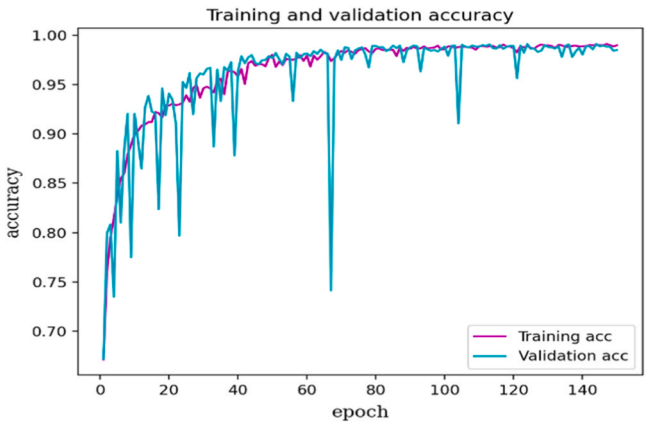



**Fig. 7.** Accuracy curve for the proposed model on MFPT dataset.

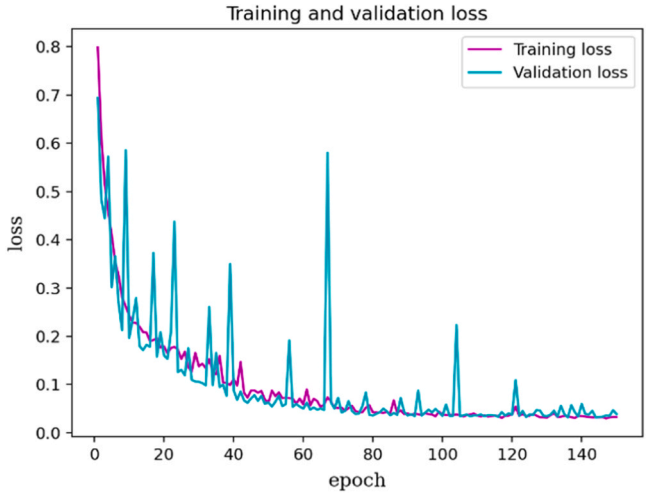



**Fig. 8.** Loss curve for the proposed model on MFPT data.

#### 4.2.3. Ablation study

Ablation study was conducted to concentrate on the impact of MHA,

**Table 4**
Performance details of the proposed model on MFPT data.

| Model | Average accuracy (%) | Number of parameters | Training time (s) | Inference time per sample | Memory usage |
|---|---|---|---|---|---|
| MHA-T | 97.11 | 300099 | 2352 | 1.76 ms | 2102.87 MB |

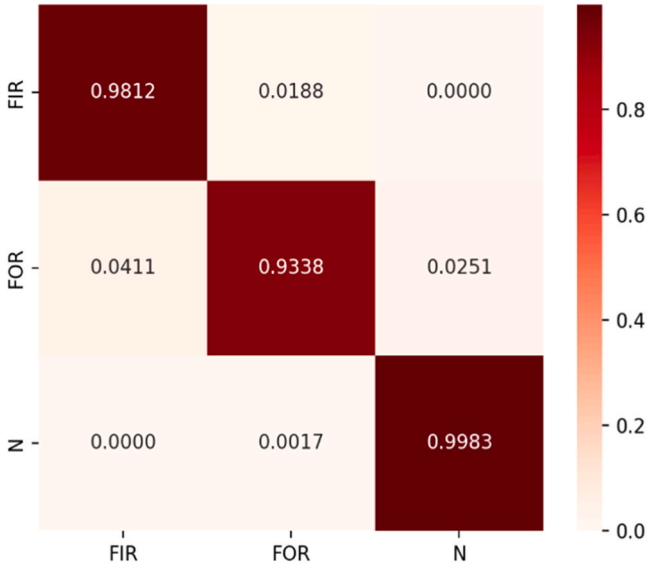



**Fig. 9.** CMX of the proposed model on MFPT data.

**Table 5**
Performance factors of the proposed model on MFPT data.

| State | *p* | *s* | F1 score |
|---|---|---|---|
| FIR | 0.9598 | 0.9812 | 0.9704 |
| FOR | 0.9785 | 0.9338 | 0.9556 |
| N | 0.9755 | 0.9983 | 0.9868 |

the number of attention heads, and the segment length of the input sequence in order to further support the proposed structure. These elements are essential to transformer-based model design, and evaluating their role clarifies the reasoning behind the configuration that was selected. The contribution of the MHA, the number of attention heads, and the length of the input segment to the overall model performance were compared. Accuracy consistently decreased when MHA was eliminated and a straight skip connection was used in its place, indicating that cross-segment feature interaction is essential for fault classification. The results demonstrated that performance improved as the number of attention heads climbed up to four, whereas subsequent increases had insignificant or even negative impacts due to overparameterization. The number of attention heads was varied between two and four. Window sizes of 256, 512, and 784 samples were used in order to investigate the impact of segment length. Longer segments improved performance by capturing richer temporal dynamics, whereas shorter segments produced lower accuracy due to insufficient temporal context; the 512–784 range produced the greatest results. The comprehensive ablation findings are shown in Table 6. Overall, the ablation study validates the architectural decisions in the proposed structure by showing that the most efficient configuration is achieved by enabling MHA, employing four attention heads, and implementing moderate segment lengths.

**Table 6**
Ablation study based on MHA, number of heads and segment length on MFPT dataset.

| MHA | Heads | Segment length | Accuracy (%) |
|---|---|---|---|
| TRUE | 2 | 256 | 68.24 |
| | 2 | 512 | 74.45 |
| | 2 | 784 | 82.38 |
| | 4 | 256 | 76.65 |
| | 4 | 512 | 91.43 |
| | 4 | 784 | 97.11 |
| FALSE | 2 | 256 | 65.45 |
| | 2 | 512 | 70.34 |
| | 2 | 784 | 75.58 |
| | 4 | 256 | 77.56 |
| | 4 | 512 | 81.24 |
| | 4 | 784 | 89.46 |

#### 4.2.4. Performance evaluation in noisy environments

Analyzing the model's performance in a noisy setting is important. An approach that is widely used to simulate noisy environments is the addition of white Gaussian noise, which is used to further analyze the effectiveness of the suggested BFD technique. Adding noise serves the main purpose of assessing the proposed method's efficacy and confirming its robustness in handling noisy data. It also helps comprehend the algorithms' limitations and provides information about how well they perform in challenging settings. The SNR range values used in this study have been thoroughly studied in the field of FD. For the analysis, SNR range ranges between −5 and 10 dB have been considered. Multiple independent runs of each experiment are carried out, keeping the parameters and structure of the proposed model constant. Also, to extend robustness of the proposed approach, Laplace noise was also injected, which better represents heavy-tailored industrial disturbances. This provides a more realistic robustness evaluation. Fig. 10 shows the accuracy of the proposed model under different SNR levels for both noises. The results demonstrate that the proposed model's classification accuracy often increases as SNR levels rise. Classification is made easier by higher SNR, which denotes less noise in the input signal. The model's performance varies with the SNR level. Overall, it is clear that although noisy situations can impact the model's performance, the proposed approach has also proven effective in noisy environments.

### 4.3. Cross-domain generalization study

To evaluate the generalization capabilities of the proposed approach, a cross-domain generalization study is done using the CWRU and MFPT datasets. Cross-domain evaluation evaluates the robustness of the learned representations under distribution shift triggered by variations in operating conditions, sensor configurations, fault severities, and data acquisition systems, in contrast to within-domain experiments where training and testing samples come from the same dataset. Two experimental scenarios were considered: training on CWRU and testing on MFPT (CWRU → MFPT) and training on MFPT and testing on CWRU (MFPT → CWRU). Without any domain adaptation or fine-tuning, the model was assessed on the target dataset after being trained solely on the source dataset in both cases. Both datasets were subjected to the same preprocessing and segmentation parameters (window length = 784, stride = 300) and a single label encoding approach to ensure fairness. Table 7 summarizes the within-domain and cross-domain results. It is evident from Table 7 that the within-domain performance is

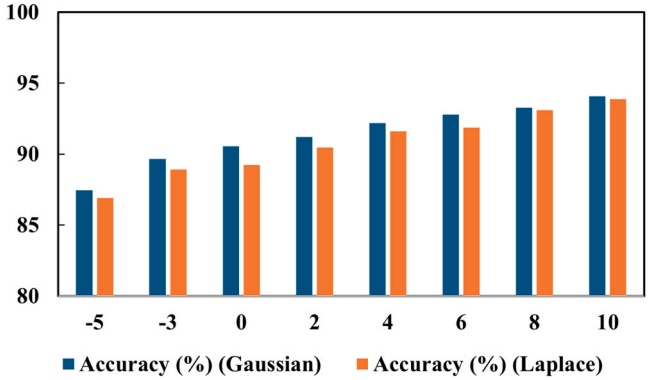



**Fig. 10.** Performance of the proposed method under different SNRs on MFPT data.

**Table 7**
Cross-domain generalization study on CWRU and MFPT datasets.

| Train dataset | Test dataset | Accuracy (%) |
|---|---|---|
| CWRU | CWRU | 99.52 |
| MFPT | MFPT | 97.11 |
| CWRU | MFPT (CWRU → MFPT) | 76.16 |
| MFPT | CWRU (MFPT → CWRU) | 72.45 |

substantially better than the cross-domain performance. There is a domain shift between the two datasets, as evidenced by the degradation seen during cross-dataset evaluation. Many factors, such as variations in sampling frequency, load conditions, fault sizes, sensor positioning, and background noise characteristics, might be attributed to this performance disparity. Variations in signal distribution limit the direct transferability of the proposed model across domains since it learns dataset-specific spectral and temporal patterns. However, the proposed model has reasonable performance in the cross-domain study, which demonstrates its efficacy.

### 4.4. Comparison with state-of-the-art methods

The above performance evaluation of the developed method demonstrates the model performance under noisy environments. The primary focus of these comparisons was the performance play of the network over various operational conditions. However, a comparison with the other BFD methods is crucial to verifying the efficacy of the developed approach. This section compares the other baseline methods that have employed comparable datasets for the BFD. These comparison methods include the deep wavelet AE (DWAE) [26], AE-based deep neural network (AE−DNN) [27], multi-scale deep CNN (MSD−CNN) [28], hierarchical deep belief network (HDBN) [29], generative adversarial network (GAN) model [30], deep recurrent neural network model (DRNN) [31], weighted principal component analysis with the Gaussian mixture model (WPCA−GMM) [32], and a combination of wavelet packet decomposition and the DBN model (WPD−DBN) [33]. A comprehensive performance background for the proposed transformer-based framework is provided by the comparison results shown in Table 8, which are taken from the corresponding original publications. It should be mentioned that these investigations were carried out in a variety of experimental conditions, such as differing hardware configurations, training methods, segmentation techniques, and data preparation. As a result, rather than being rigorously controlled benchmarking research, the comparison is meant to serve as a general performance reference. The ablation analysis of the proposed model under a unified protocol is separately reported in previous sections to ensure fair internal evaluation. The table clearly shows that the proposed method achieved performance comparable to state-of-the-art approaches in fault classification accuracy across different bearing fault types.

**Table 8**
Comparison of the proposed method with the state-of-the-art methods based on the CWRU dataset.

| **Method [Ref.]** | **Accuracy (%)** |
|---|---|
| DWAE [26] | 95.20 |
| AE-DNN [27] | 97.47 |
| MSD-CNN [28] | 99.27 |
| HDBN [29] | 99.03 |
| GAN [30] | 96 |
| DRNN [31] | 98.67 |
| WPCA-GMM [32] | 82.50 |
| WPD-DBN [33] | 98.24 |
| MHA-T (CWRU) | 99.52 |

### 4.5. Discussion and future challenges

The proposed method has used transformers with multi-head attention mechanism for the BFD. The model was evaluated on the two bearing dataset to understand the performance on different bearing datasets. The proposed model performed efficiently on both the datasets, where it achieved consistent and competitive results. This shows the model's capability to generalize across datasets collected under diverse conditions. But even with these positive outcomes, there are still limitations. For instance, data from machinery with different structural configurations, sensor locations, or operational loads may be used in real-world applications, which could nevertheless have an impact on the model's transferability. Furthermore, even though the model performed well in noisy settings, further extensive validation on a variety of industrial datasets is required.

These constraints could be addressed in future research using a variety of approaches. It may be possible to effectively adapt the model to unknown datasets by using domain adaptation and transfer learning techniques. Expanding the range of possibilities in training circumstances may be facilitated by physics-informed simulation and data augmentation. Additionally, by utilizing several complementary models, ensemble learning techniques may increase robustness. Lastly, thorough cross-domain benchmarking across various pieces of equipment and operating conditions would offer more profound understanding of the proposed approach's scalability and industrial suitability.

## 5. Conclusion

In this study, we investigated the application of a transformer with a multi-head attention mechanism for bearing fault detection on the CWRU and MFPT bearing dataset. The experimental analysis displayed the efficacy of the developed method in identifying suitable fault patterns and achieving high classification performance. One of the major advantages of the proposed method lies in its ability to automatically learn important representations from the raw vibration signal data, without depending on handcrafted features or domain-specific knowledge. Originally intended for sequence modeling challenges in natural language processing, the transformer architecture has shown itself to be a good fit for capturing the complex connections and temporal dependencies observed in vibration signals. Using an MHA mechanism, the model has categorized faults in an efficient manner by paying attention to relevant fragments of the input sequence. Also, the performance of the proposed model was evaluated under noisy environmental conditions and performed reasonably well in these conditions. The comparison showed that the proposed method achieved performance comparable to state-of-the-art approaches in fault classification accuracy across different bearing fault types. To sum up, the amalgamation of the transformer and multi-head attention mechanism represents a noteworthy progression in the domain of bearing fault detection, offering an efficient data-driven methodology for rotating machinery health monitoring and predictive maintenance in industrial environments.

## CRediT authorship contribution statement

**Muhammad Muzammil Azad:** Writing – review & editing, Investigation, Formal analysis, Conceptualization. **Heung Soo Kim:** Writing – review & editing, Validation, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Prashant Kumar:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Conceptualization. **Izaz Raouf:** Writing – review & editing, Investigation, Formal analysis, Conceptualization.

## Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Data will be made available on request.

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